RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2023-24) -- SEMESTER 4

COURSE: THEORY OF AUTOMATA AND FORMAL LANGUAGES (TAFL) (CS-402)

Time Allowed: 3 Hours Maximum Marks: 100 Marks Paper Code: CS-402

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Define DFA formally as a 5-tuple (Q, Sigma, delta, q0, F).

Q1. (b) What is difference between DFA and NFA in terms of transition function?

Q1. (c) State Arden theorem for solving regular expression equations.

Q1. (d) State Pumping Lemma for Regular Languages.

Q1. (e) Define ambiguous context-free grammar with an example.

Q1. (f) What is Chomsky Normal Form (CNF) and Greibach Normal Form (GNF)?

Q1. (g) Differentiate between DPDA (Deterministic PDA) and NPDA.

Q1. (h) Define Turing Machine as a 7-tuple.

Q1. (i) What is Halting Problem of Turing Machine? Is it decidable?

Q1. (j) Define Chomsky Hierarchy of formal grammars (Type 0, 1, 2, 3).

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Convert given NFA with epsilon transitions into an equivalent DFA using subset construction.

Q2. (b) Minimize given DFA with 6 states using Table Filling / Myhill-Nerode equivalence method.

Q2. (c) Using Pumping Lemma, prove that language L = {a^n b^n | n >= 0} is not regular.

Q2. (d) Construct a Pushdown Automaton (PDA) accepting language L = {w c w^R | w in {a, b}*} by empty stack.

Q2. (e) Design a Turing Machine that accepts the language L = {a^n b^n c^n | n >= 1} with state transition table.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Construct DFA accepting strings over {0, 1} having even number of 0s and odd number of 1s. [10 Marks]

-- OR --

(b) Find regular expression for language over {a, b} containing substring 'ab' but not 'ba'. [10 Marks]

Q4. (a) Convert CFG into Chomsky Normal Form (CNF): S -> aSb | ab | epsilon, S -> AB, A -> aA | a, B -> bB | b. [10 Marks]

-- OR --

(b) Remove null productions, unit productions, and useless symbols from given grammar. [10 Marks]

Q5. (a) Construct PDA for language L = {a^n b^(2n) | n >= 1} with transition diagram and trace. [10 Marks]

-- OR --

(b) Convert Context-Free Grammar into Pushdown Automaton using standard algorithm. [10 Marks]

Q6. (a) Design a Turing Machine that computes 2's complement of an arbitrary binary string. [10 Marks]

-- OR --

(b) Design a Turing Machine for unary addition f(m, n) = m + n. [10 Marks]

Q7. (a) Prove that Post Correspondence Problem (PCP) is undecidable. [10 Marks]

-- OR --

(b) Explain Rice Theorem and prove that non-trivial semantic properties of programs are undecidable. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****